CASE STUDY #3 — TELECOM

FastConnect Telecom's 5G Rollout Crisis

Network Expansion & Service Assurance

Introduction

FastConnect Telecom, a mid-sized mobile network operator (MNO) with 8 million subscribers, launched an aggressive 5G rollout across 200 cities. Within three months of launch, the company began receiving a flood of customer complaints about dropped calls, slow data speeds, and service outages in supposedly "covered" areas.

Background

Field engineers were manually cross-checking spreadsheets to determine which towers were live, under maintenance, or faulty — a process taking days instead of hours. FastConnect's OSS stack had not been scaled or integrated properly to support the new 5G infrastructure:

Network inventory management was outdated and did not reflect real-time status of newly deployed 5G small cells and core network elements. Fault management and network monitoring systems were still tuned for 4G thresholds, so many 5G-specific faults went undetected or were misclassified. Service provisioning workflows were partly manual, and trouble ticketing was not integrated with the Network Operations Center (NOC).

Approach: OSS Concepts Applied

OSS Function	Issue Identified	Fix Implemented
Network Inventory (physical & logical)	Inventory database out of sync with real deployments	Automated inventory reconciliation using network discovery tools
Fault Management (FCAPS – Fault)	Alarms not correlated; alarm flooding	Deployed root-cause alarm correlation engine
Performance Management (FCAPS – Performance)	No visibility into 5G-specific KPIs (latency, throughput per slice)	Introduced 5G-specific performance dashboards
Service Order Management	Manual provisioning delays	Automated zero-touch provisioning (ZTP) for new cell sites
Trouble Ticketing / Assurance	Disconnected from BSS-side customer complaints	Integrated ticketing system linking CRM complaints to NOC alarms

Resolution & Outcome

FastConnect implemented an integrated OSS platform with real-time network inventory, automated fault correlation, and closed-loop assurance, where a customer complaint automatically cross-references live network alarms in the same geography. Within four months:

- Mean Time to Repair (MTTR) dropped by 45%
- Customer complaint volume for network issues fell by 30%
- Tower activation time reduced from 3 days to 6 hours

Final Note

The crisis was not a coverage problem but a systems-integration problem: FastConnect's OSS stack had not scaled with its network. Fixing the visibility and automation gap — not adding more towers — resolved the customer experience issue.

Questions for Discussion

- Why can an aggressive rollout succeed at the RF/coverage level and still fail at the customer-experience level?

- What risks does closed-loop assurance (auto-linking complaints to alarms) introduce, and how should they be managed?
- How should FastConnect prioritize which OSS gaps to fix first when multiple systems are broken simultaneously?